RiskShield-Hierarchical HRMPPO-MPC: Causal Hierarchical Safe Reinforcement Learning for Construction-Site Inspection under Perception Uncertainty

Mahmoud Maher

***Abstract*—Autonomous construction inspection couples sparse coverage objectives with collisions, worker proximity, restricted areas, and imperfect visual perception. We present a reproducible Gymnasium benchmark, RiskShield-PPO (PPO-Lagrangian with an auxiliary cost-value estimator), and a predictive action shield. A five-algorithm matrix evaluates 1,350 episodes over three site sizes, three hazard densities, three perception-noise levels, and ten paired seeds; six ablations add 180 episodes. RiskShield-PPO reduced mean safety cost from PPO's 49.39 to 17.13 and eliminated observed collisions, but neither learned method completed a full mission. Frontier exploration achieved 96.67% success and 0.998 hazard recall, exposing a substantial learning gap. A separate Webots runtime verifies CUDA perception, shielded policy-to-motor control, and rendered Webots operation. The project also evaluates an experimental hierarchical v4 policy on 270 paired held-out grid episodes. It achieved 58.89% mission success and 92.14% hazard recall in that protocol; production remains RiskShield-PPO v1 because the v4 replacement gates were rejected. The evidence is simulation-only and does not establish real-world safety.**

I. Introduction

Construction inspection requires visiting hazards while avoiding workers, restricted regions, machinery, and fall risks. Reward maximization alone does not express an acceptable risk budget, and perception errors make fixed safety assumptions unreliable. Safe RL commonly modifies the objective or constrains exploration [1]; constrained policy optimization formalizes cost limits [2], while shields can replace actions before execution [3].

This project contributes (i) a typed, deterministic construction-inspection benchmark; (ii) genuine PPO, SAC, planner, and constrained baselines; (iii) RiskShield-PPO and a separable k-step shield; (iv) controlled perception uncertainty; and (v) an integrated Webots demonstration. We make no claim of learned-policy task mastery: the complete matrix shows zero mission success for all three learned policies.

II. Related Work

PPO optimizes a clipped policy-ratio surrogate [4]. SAC is an off-policy maximum-entropy actor–critic method [5]. CMDP methods optimize reward subject to expected cost [2]. Shielding instead filters proposed actions with an external safety model [3]. Our system combines a Lagrangian training signal with a predictive runtime filter; it does not inherit formal guarantees from either cited method. Gymnasium provides the standardized environment API [6], and Webots supplies the robot simulator [7].

III. Problem Formulation

We model a finite-horizon CMDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, c, \gamma, d)$. The discrete actions are north, south, west, east, and inspect. The observation contains padded semantic map channels (obstacle, hazard, worker, restricted, visited, agent, risk, dynamic hazard, PPE risk, visibility), pose, orientation, progress, local risk, and an action mask.

Reward separates task progress from cost:

$$r_t = -0.01 + r_{\text{inspect}} + r_{\text{coverage}} - r_{\text{invalid}}, \tag{1}$$

while

$$c_t = c_{\text{collision}} + c_{\text{near}} + c_{\text{restricted}} + c_{\text{PPE}}. \tag{2}$$

The objective is

$$\max_{\pi} J_R(\pi) \quad \text{s.t.} \quad J_C(\pi) \leq d, \tag{3}$$

with training budget $d = 5$. Reward and cost remain separately logged.

IV. Construction Safety RL Benchmark

Three parameterized grid sites vary size, hazards, workers, dynamic obstacles, and episode horizon. Reset and transition randomness use explicit seeds. Obstacle-aware A*, frontier exploration, and nearest-risk revisit provide interpretable expert components. The final comparison uses risk-aware A*, frontier exploration, PPO, SAC, and RiskShield-PPO.

The environment includes partial observation, semantic layers, action masks, dynamic hazards, restricted zones, PPE-risk cells, inspection targets, reward, cost, rendering, and serialized telemetry. The Gymnasium checker, deterministic reset/step, mask, collision, near-miss, cost, reward, and serialization tests pass.

V. RiskShield-PPO Method

RiskShield-PPO uses PPO's clipped objective but trains under a Lagrangian penalty:

$$\mathcal{L}(\theta, \lambda) = J_R(\pi_\theta) - \lambda \left(J_C(\pi_\theta) - d \right), \quad \lambda \geq 0. \tag{4}$$

The multiplier update is

$$\lambda_{k+1} = \Pi_{[0,100]} \left[\lambda_k + \eta_\lambda (\hat{J}_C - d) \right], \tag{5}$$

with $\eta_\lambda = 0.02$. An auxiliary neural critic minimizes

$$\mathcal{L}_C = \left(V_C(s_t) - [c_t + \gamma(1 - z_t) V_C(s_{t+1})] \right)^2. \tag{6}$$

The resulting penalized reward actually reaches PPO training. Telemetry records policy/value diagnostics, cost-value loss, cost, budget, and multiplier. Training stopped after 15,300 steps without improvement; the multiplier saturated, a negative result reported rather than hidden.

VI. Hierarchical Experimental System

The experimental v4 system separates learned mission decisions from local execution. A recurrent option policy selects exploration, inspection, retreat, and replanning intentions and a target in the causal observed map. An incremental risk-aware planner proposes a local route; a predictive controller and Safety Shield validate the resulting primitive before wheel commands. The policy never receives hidden map state or future worker motion. This is not end-to-end wheel-velocity reinforcement learning. H3 reached 61.54% success with 95.51% recall; H4 imitation reached 62.50% success with mean cost 16.31, while H4 30k fell to 56.25%. These negative results motivate retaining v1 in production.

VII. Predictive Safety Shield

For proposed action a_t, the shield rolls forward a deterministic local model for horizon k. It predicts collisions, near misses, restricted entry, proximity, and budget excess. It accepts, replaces, masks, or stops the action.

Algorithm 1 Predictive shield

Receive s_t, proposed a_t, budget b_t
for a in proposed-first candidate order **do**
    $\tau, \hat{C} \leftarrow \mathrm{Rollout}(s_t, a, k)$
    if τ has no violation and $\hat{C} \leq b_t$ **then**
        return structured decision with final action a
    end if
end for
return emergency stop with "no safe action"

The structured record includes proposed and final actions, trajectory, cost, violations, decision, confidence, emergency state, and latency. In Webots, the final shielded action, not the proposal, reaches wheel commands.

VIII. Perception and Semantic Risk Mapping

The integrated path is Webots camera $\rightarrow$ CUDA detector $\rightarrow$ typed detections $\rightarrow$ temporal filtering $\rightarrow$ semantic risk state $\rightarrow$ policy and shield. The production checkpoint has 14 declared classes and a recorded SHA-256 hash. Unsupported concepts use explicitly labeled simulator ground truth rather than fabricated detections. The corrected hierarchical Webots runtime records 100 policy decisions per world with real camera timestamps, recurrent-state hashes, target persistence, planner/controller/shield traces, and wheel commands. In the tested worlds the detector produced no valid detections, so the v4 report truthfully leaves perception-driven semantic change false; no CV metric is fabricated.

IX. Experimental Setup

The primary matrix is 3 site sizes $\times 3$ hazard densities $\times 3$ false-negative probabilities $\times 5$ algorithms $\times 10$ paired evaluation seeds, totaling 1,350 episodes. Cache records include immutable configuration and checkpoint hashes. Planner episodes run concurrently; GPU policy inference is serialized. No missing run is imputed.

We report mean, standard deviation, median, IQR, and 95% t intervals per cell. Friedman tests compare all algorithms because bounded, zero-inflated metrics need not be normal. A planned paired Wilcoxon comparison contrasts RiskShield-PPO and PPO, with Holm correction over six metrics. Standardized paired mean difference is the effect size.

The corrected Webots integration was validated headlessly in the small, medium, and dynamic worlds. Each run reached 100 hierarchical decisions and used the H4 target-persistence checkpoint, while RiskShield-PPO v1 remained the production policy. The earlier three-step adapter is retained only as superseded startup evidence. A visible real-time demo was attempted with the same controller; desktop Qt startup is environment-dependent and is reported separately from the accepted headless runtime evidence.

X. Main Results

RiskShield-PPO improved PPO's hazard recall (0.147 versus 0.083), coverage (0.0275 versus 0.0090), and safety cost (17.13 versus 49.39), with no observed collision. SAC reached 0.074 coverage but incurred 150.86 mean safety cost and a 0.796 collision rate. Neither learned policy completed a mission. Frontier exploration achieved 0.967 success and 0.998 recall; risk-aware A* reached 1.0 success and recall but lower coverage.

XI. Uncertainty Experiments

False-negative probabilities are 0, 0.1, 0.2, and 0.3 in the dedicated uncertainty suite. Lighting, blur, camera noise, moving-obstacle density, and unseen layout are seeded. Clean truth, perturbed perception, and observation are separate. Across 320 episodes and 64 conditions, mean robustness was 0.8893. Because baseline learned-policy mission success was zero, its ratio is neutral rather than replaced by an invented improvement.

TABLE I
PRIMARY BENCHMARK MEANS OVER 270 EPISODES PER ALGORITHM.

Algorithm	Runs	Recall	Coverage	Collision rate	Safety cost	Success	Robustness
PPO	270	0.0834	0.0090	0.0741	49.3852	0.0000	0.0962
RiskShield-PPO	270	0.1466	0.0275	0.0000	17.1296	0.0000	0.1624
SAC	270	0.1339	0.0744	0.7963	150.8556	0.0000	0.0974
Frontier exploration	270	0.9978	0.5487	0.0000	4.0111	0.9667	0.6935
Risk-aware A*	270	1.0000	0.1976	0.0000	2.2500	1.0000	0.6085

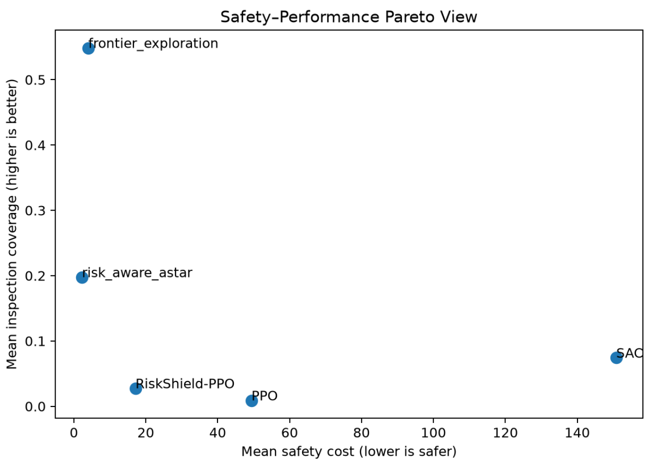

Fig. 1. Safety cost versus inspection coverage from raw primary records.

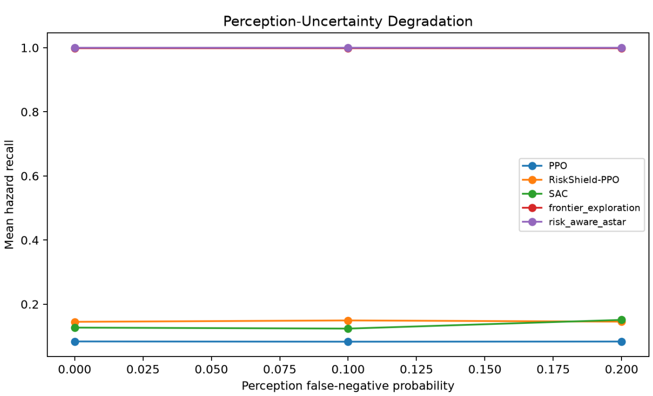

Fig. 2. Hazard-recall degradation derived from primary raw results.

XII. ABLATION STUDY

Removing the predictive shield raised mean safety cost from 8.43 to 66.52. The one-step shield produced lower cost (2.07) than $k = 3$ in this limited sample, so longer prediction was not uniformly better. Removing CV risk injection or replacing the constrained checkpoint with PPO yielded identical reported task means under the shield, although their internal actions and intervention counts differed. This indicates that the shield dominated these weak policies. Removing action masking reduced coverage.

XIII. DISCUSSION

The planners' advantage indicates that exploration and long-horizon credit assignment, not only collision avoidance, remain unsolved. RiskShield-PPO's lower cost is meaningful but insufficient: a stationary or repeatedly corrected policy can be safe and useless. The saturated multiplier suggests the selected budget was difficult relative to the training distribution. Future work should increase training budgets, tune cost scaling, evaluate recurrent policies only after training a valid checkpoint, and separate shield dependence from intrinsic policy safety.

XIV. LIMITATIONS

All research evaluation is simulation-only. Learned policies had zero complete mission success. Stage 5C is a bounded integrated demonstration, not a long-duration deployment. The detector supports only its recorded class list; other semantic layers use simulator truth. No recurrent checkpoint exists, so no recurrent ablation is claimed. Results cover one laptop GPU and Webots R2025a. The shield is model-based and can fail under unmodeled dynamics. No real-world safety guarantee, certification, or human-subject evaluation is provided.

XV. ETHICAL AND SAFETY CONSIDERATIONS

The system must not replace qualified safety personnel or authorize entry into hazardous areas. Simulator performance can create false confidence. Deployment would require site-specific risk assessment, fail-safe hardware, independent verification, privacy controls, worker consultation, and compliance with local law and safety standards.

XVI. CONCLUSION

This work provides a reproducible construction Safe RL benchmark and truthful evidence spanning planning, learned policies, constrained optimization, shielding, uncertainty, CUDA perception, and Webots motor control. RiskShield-PPO reduced cost but did not solve inspection; expert planners remained substantially stronger. The negative result defines a clear research target rather than a deployment claim.

REPRODUCIBILITY STATEMENT

The repository records configuration files, seeds, checkpoint hashes, atomic per-run caches, raw CSV and Parquet data, statistical scripts, figure source data, tests, and PowerShell entry points. The complete benchmark command resumes only

TABLE II

SIX PAIRED ABLATIONS; 30 EPISODES PER CONFIGURATION.

Variant	Episodes	Recall	Coverage	Safety cost	Collision rate	Success
One-step shield	30	0.1661	0.0569	2.0667	0.0000	0.0000
Full RiskShield-PPO	30	0.1522	0.0408	8.4333	0.0000	0.0000
Without action masking	30	0.1211	0.0206	5.9000	0.0000	0.0000
Without cost objective	30	0.1522	0.0408	8.4333	0.0000	0.0000
Without CV risk	30	0.1522	0.0408	8.4333	0.0000	0.0000
Without predictive shield	30	0.1189	0.0291	66.5167	0.0000	0.0000

missing runs. The production command separately reproduces the Webots integration evidence.

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